

Block-level FRQI preparation (structural view)

Registers: m position qubits + 1 color qubit (basis $|addr, c\rangle$)

$|0\rangle^{\otimes(m+1)}$

For each pixel $p = 0 \dots N_{\text{pix}} - 1$

Address
compare

ctrl- $R_y(\theta_p)$

Uncompute
compare

v-chain uses $\leq m - 2$ ancilla qubits (reused each iteration)

FRQI state $|\psi_{\text{FRQI}}\rangle$